

1. Math Problems

1.1 Reverse Integer

1.2 Palindrome Number

1.3 Count Digits

1.4 Factorial

1.5 Power of Number

1.6 Square Root

1.7 Trailing Zeroes

1.8 Sum of Digits

1.9 Armstrong Number

1.10 Perfect Number

2. Number Theory + Special Numbers

- 2.1 GCD
- 2.2 LCM
- 2.3 Prime Check
- 2.4 Sieve of Eratosthenes
- 2.5 Modular Exponentiation
- 2.6 Prime Factors
- 2.7 Divisors Count
- 2.8 Euler Totient
- 2.9 Co-prime
- 2.10 Extended Euclid
- 2.11 Armstrong Number
- 2.12 Strong Number
- 2.13 Perfect Number
- 2.14 Happy Number
- 2.15 Neon Number
- 2.16 Automorphic Number
- 2.17 Palindrome Number
- 2.18 Harshad Number
- 2.19 Duck Number
- 2.20 Spy Number

3. Bit Manipulation

3.1 Single Number

3.2 Count Set Bits

3.3 Power of Two

3.4 Missing Number

3.5 XOR till N

3.6 Swap using XOR

3.7 Rightmost Set Bit

3.8 Subsets

3.9 Bit Difference

3.10 Divide without operator

4. 1D Arrays

4.1 Linear Search

4.2 Largest Element

4.3 Second Largest

4.4 Rotate Array

4.5 Move Zeros

4.6 Remove Duplicates

4.7 Missing Number

4.8 Max Consecutive Ones

4.9 Kadane Algorithm

4.10 Stock Buy Sell

5. Strings

5.1 Reverse String

5.2 Palindrome String

5.3 Anagram

5.4 Longest Common Prefix

5.5 String to Integer

5.6 Count Vowels

5.7 Remove Duplicates

5.8 Substring Search

5.9 Frequency Count

5.10 Roman to Integer

6. 2D Arrays

6.1 Matrix Transpose

6.2 Spiral Traversal

6.3 Row Sum

6.4 Search Matrix

6.5 Rotate Matrix

6.6 Diagonal Sum

6.7 Wave Traversal

6.8 Boundary Traversal

6.9 Set Matrix Zero

6.10 Island Count

7. Searching

7.1 Linear Search

7.2 Binary Search

7.3 First Occurrence

7.4 Last Occurrence

7.5 Floor and Ceil

7.6 Search Insert Position

7.7 Peak Element

7.8 Rotated Array Search

7.9 Count Occurrences

7.10 Square Root

8. Sorting

8.1 Bubble Sort

8.2 Selection Sort

8.3 Insertion Sort

8.4 Merge Sort

8.5 Quick Sort

8.6 Count Sort

8.7 Sort Colors

8.8 Kth Largest

8.9 Relative Sort

8.10 Frequency Sort

9. Binary Search Advanced

9.1 Classic Binary Search

9.2 First Bad Version

9.3 Peak Element

9.4 Search in Rotated Array

9.5 Minimum in Rotated Array

9.6 Allocate Pages

9.7 Aggressive Cows

9.8 Painter Partition

9.9 Koko Bananas

9.10 Median of Two Sorted Arrays

10. Two Pointers

10.1 Pair Sum

10.2 Remove Duplicates

10.3 Container With Water

10.4 3 Sum

10.5 4 Sum

10.6 Merge Sorted Arrays

10.7 Reverse Vowels

10.8 Move Zeros

10.9 Sort 0 1 2

10.10 Valid Palindrome

11. Sliding Window

- 11.1 Max Sum Subarray (K)
- 11.2 Longest Substring Without Repeat
- 11.3 Minimum Window Substring
- 11.4 Fruits Basket
- 11.5 Max Consecutive Ones III
- 11.6 Anagram Count
- 11.7 Subarray Product < K
- 11.8 Longest Repeating Replacement
- 11.9 Binary Subarray Sum
- 11.10 Sliding Window Maximum

12. Prefix Sum

12.1 Prefix Sum Array

12.2 Range Sum Query

12.3 Subarray Sum = K

12.4 Equilibrium Index

12.5 Product Except Self

12.6 Pivot Index

12.7 2D Prefix Sum

12.8 XOR Prefix

12.9 Subarray Divisible by K

12.10 Count Subarrays

13. Kadane Variations

13.1 Maximum Subarray

13.2 Circular Subarray

13.3 Maximum Product Subarray

13.4 Minimum Subarray

13.5 Largest Sum K

13.6 Maximum Difference

13.7 Rainwater Trapping

13.8 Max Sum Rectangle

13.9 Stock II

13.10 Max Absolute Sum

14. Hashing

14.1 Frequency Count

14.2 Two Sum

14.3 Intersection of Arrays

14.4 Union of Arrays

14.5 Longest Subarray Sum = K

14.6 Subarray with 0 Sum

14.7 Count Distinct Elements

14.8 First Repeating Element

14.9 Longest Consecutive Sequence

14.10 Zero Sum Subarrays

15. HashMap / TreeMap

15.1 Two Sum

15.2 LRU Cache

15.3 Top K Frequent Elements

15.4 Group Anagrams

15.5 Subarray Sum = K

15.6 First Unique Character

15.7 Isomorphic Strings

15.8 Word Pattern

15.9 Frequency Sort

15.10 Map Operations

16. HashSet / TreeSet

- 16.1 Distinct Elements
- 16.2 Longest Consecutive Sequence
- 16.3 Contains Duplicate
- 16.4 Happy Number
- 16.5 Intersection
- 16.6 Union
- 16.7 Subarray Distinct Count
- 16.8 Set Operations
- 16.9 Remove Duplicates
- 16.10 Missing Number

17. Recursion

17.1 Factorial

17.2 Fibonacci

17.3 Power

17.4 Sum of Digits

17.5 Reverse Array

17.6 Palindrome Check

17.7 Tower of Hanoi

17.8 Subsequence Generation

17.9 Binary Strings

17.10 Josephus Problem

18. Backtracking

18.1 Subsets

18.2 Permutations

18.3 Combination Sum

18.4 N Queens

18.5 Sudoku Solver

18.6 Rat in Maze

18.7 Word Search

18.8 Palindrome Partition

18.9 Generate Parentheses

18.10 Combination Sum II

19. Linked List

19.1 Reverse Linked List

19.2 Detect Cycle

19.3 Middle Node

19.4 Merge Two Lists

19.5 Remove Nth Node

19.6 Intersection Point

19.7 Palindrome Linked List

19.8 Add Two Numbers

19.9 Delete Node

19.10 Sort Linked List

20. Stack

20.1 Valid Parentheses

20.2 Next Greater Element

20.3 Stock Span

20.4 Largest Rectangle

20.5 Min Stack

20.6 Evaluate Postfix

20.7 Infix to Postfix

20.8 Reverse Stack

20.9 Celebrity Problem

20.10 Daily Temperatures

21. Queue

21.1 Implement Queue

21.2 Circular Queue

21.3 Queue Using Stack

21.4 Stack Using Queue

21.5 First Non-Repeating Character

21.6 Generate Binary Numbers

21.7 Sliding Window Maximum

21.8 Rotten Oranges

21.9 BFS Basics

21.10 Level Order Traversal

22. Deque

22.1 Sliding Window Maximum

22.2 Circular Deque Design

22.3 First Negative in Window

22.4 Max of Subarrays

22.5 Deque Operations

22.6 Sum of Min and Max

22.7 Gas Station

22.8 Shortest Subarray $\geq K$

22.9 Monotonic Queue

23. Slow and Fast Pointers

23.1 Detect Cycle

23.2 Find Middle

23.3 Happy Number

23.4 Cycle Start Node

23.5 Palindrome Linked List

23.6 Remove Nth Node

23.7 Find Duplicate Number

23.8 Intersection

23.9 Loop Length

23.10 Reorder List

24. Heap (Priority Queue)

24.1 Kth Largest Element

24.2 Top K Frequent

24.3 Merge K Sorted Arrays

24.4 Median from Stream

24.5 K Closest Points

24.6 Sort Nearly Sorted

24.7 Task Scheduler

24.8 Reorganize String

24.9 Smallest Range

24.10 Connect Ropes

25. Trees

25.1 Inorder Traversal

25.2 Preorder

25.3 Postorder

25.4 Level Order

25.5 Height

25.6 Diameter

25.7 Balanced Tree

25.8 LCA

25.9 Same Tree

25.10 Zigzag Traversal

26. Graphs

26.1 Graph Representation

26.2 BFS

26.3 DFS

26.4 Number of Islands

26.5 Cycle Detection

26.6 Bipartite Graph

26.7 Topological Sort

26.8 Shortest Path BFS

26.9 Connected Components

26.10 Graph Coloring

27. Graph Algorithms

27.1 Dijkstra

27.2 Bellman Ford

27.3 Floyd Warshall

27.4 Kruskal MST

27.5 Prim MST

27.6 Kosaraju SCC

27.7 Tarjan Bridges

27.8 Detect Cycle Directed

27.9 Shortest Path DAG

27.10 Alien Dictionary

28. Greedy

- 28.1 Activity Selection
- 28.2 Fractional Knapsack
- 28.3 Job Sequencing
- 28.4 Minimum Coins
- 28.5 Gas Station
- 28.6 Jump Game
- 28.7 Candy
- 28.8 Merge Intervals
- 28.9 Non Overlapping Intervals
- 28.10 Assign Cookies

29. DP (Arrays)

29.1 Fibonacci DP

29.2 Climbing Stairs

29.3 House Robber

29.4 Max Sum Non Adjacent

29.5 LIS

29.6 Coin Change

29.7 Partition Equal Subset

29.8 Subset Sum

29.9 Min Cost Climbing

29.10 Decode Ways

30. DP (Matrix)

30.1 Unique Paths

30.2 Minimum Path Sum

30.3 Maximum Square

30.4 Cherry Pickup

30.5 Dungeon Game

30.6 Triangle Path

30.7 Falling Path Sum

30.8 Grid Paths with Obstacles

30.9 Largest Rectangle Matrix

30.10 Count Squares